

Video-by-Video Analysis of AI Engineer RAG Talks for Your Vector-Free Retrieval Dissertation

TL;DR

- **The four most citable talks are the ones grounded in peer-reviewed work:** NVIDIA's HybridRAG (Sarmah et al., arXiv:2408.04948), Microsoft's GraphRAG (Jonathan Larson; Edge et al., arXiv:2404.16130), Vectara's open-rag-eval (UMBRELA/AutoNuggetizer, built with Jimmy Lin's U. Waterloo lab), and Mixedbread's agentic-retrieval talk — the rest are practitioner corroboration or motivating anecdotes, and a few are too promotional/domain-mismatched to cite.
- **Your central novelty claim survives scrutiny:** across the AI Engineer 2023-2026 archives there is NO talk whose thesis is query-adaptive retrieval-*paradigm* routing grounded in Adaptive-RAG/Self-RAG/CRAG/FLARE — the closest talks (Bergum, Lichtenberg) frame adaptivity as *agentic tool-use*, not a trained per-query strategy router. This supports rather than threatens your differentiation.
- **Multiple independent practitioner talks corroborate H1-H3 and H5** (BM25's surprising strength, graph retrieval for entity/multi-hop, and the retrieval-vs-generation faithfulness gap), giving you industry triangulation to cite alongside academic sources.

Key Findings

1. The academic-grounding split is the single most useful lens for citing these videos. Four talks trace directly to published research and can be cited as industry implementations of peer-reviewed methods. The graph-RAG cluster (NVIDIA HybridRAG, Microsoft GraphRAG, Writer) collectively corroborates H3. The evaluation cluster (Vectara, Quotient AI, Qdrant) corroborates H5 and your decision to separate retrieval metrics from generation faithfulness. The lexical-comeback cluster (Elastic ×2, Mixedbread, Turbopuffer) corroborates H1.

2. Your novelty gap is confirmed by a negative finding. No AI Engineer talk presents a *trained* per-query router choosing lexical vs dense vs graph, and none cites the academic routing canon. Conference discourse treats retrieval-method choice as either static hybrid (always BM25+vector+RRF+rerank) or agentic/iterative tool use.

3. Practitioner consensus has swung toward "hybrid-by-default + rerank," not routing. Important for your Framework chapter: industry defaults to always-on fusion, whereas your contribution is a principled per-query paradigm-selection framework — a genuinely different design point you should position explicitly. (Multiple 2026 conference recaps note "RAG is fading from the center... it's been absorbed" into agentic systems — useful context for why routing-as-you-define-it is under-explored on stage.)

Details: Video-by-Video Analysis

1. "Information Retrieval from the Ground Up" — Philipp Krenn, Elastic

- *Speaker/affiliation/event*: Philipp Krenn, developer-relations/search advocate at Elastic; AI Engineer World's Fair (Search & Retrieval track), workshop-style session.
- *Content*: Thesis (verbatim from description): "Vector search is only a feature. Search engines and information retrieval have retaken their position as the foundation of RAG." Walks IR fundamentals: BM25 as lexical baseline (TF-IDF extended with term-frequency saturation and length normalization), dense kNN/HNSW for semantic queries, and hybrid search fusing BM25 + vector via Reciprocal Rank Fusion (RRF). Krenn is a recognized RRF popularizer (presented RRF at Haystack 2023).
- *Maps to*: H1 (lexical competitiveness), H4 (RRF fusion — directly relevant to your Hybrid BM25+GraphRAG RRF plan), Methodology, Literature Review.
- *Academic grounding*: Practitioner framing, but underlying techniques (BM25; RRF from Cormack, Clarke & Büttcher, SIGIR 2009) are standard published IR. Medium.
- *How to cite*: Industry corroboration for the "IR/lexical comeback" narrative and a practitioner reference for RRF as the pragmatic fusion default; pair with the Cormack et al. RRF paper for the method itself.
- *Quote to extract*: "Vector search is only a feature."

2. "Layering Every Technique in RAG, One Query at a Time" — David Karam, Pi Labs (fmr. Google Search)

- *Speaker/affiliation/event*: David Karam, Pi Labs (formerly Google Search); AI Engineer World's Fair.
- *Content (from description)*: Progression "from the simplest Search — in-memory embeddings with relevance ranking" up to "the most complex planet-scale Search — 70+ corpus mix of token, embedding" methods, layering techniques query-by-query.
- *Maps to*: Your Retrieval Strategy Selection Framework and Phase-2 routing extension; the "one query at a time" framing is conceptually adjacent to per-query strategy selection (though about layering, not paradigm routing).
- *Academic grounding*: Practitioner/experience-based; no explicit academic citations found.
- *How to cite*: Motivating practitioner anecdote for "different queries need different retrieval treatment"; not a citable empirical source.
- *Quote to extract*: the "70+ corpus mix of token, embedding" scaling framing.

3. "RAG Evaluation Is Broken! Here's Why—and How to Fix It" — Yuval Belfer & Niv Granot, AI21 Labs

- *Speaker/affiliation/event*: Yuval Belfer (Senior Developer Advocate) and Niv Granot, AI21 Labs; AI Engineer World's Fair 2026.
- *Content*: Argues developers over-optimize local benchmarks, chunking strategies, and "perfect retrieval scores" that don't translate to production. Companion to AI21's "Stop Chunking Like It's 2022" (Belfer), which claims fixed chunk sizes are a performance killer and a multi-index approach can deliver up to 37% better recall with no retraining.
- *Maps to*: H5 (retrieval metrics $\neq$ generation quality — essentially an industry statement of H5), Evaluation, Limitations.
- *Academic grounding*: Practitioner opinion; no peer-reviewed citations found.
- *How to cite*: Industry corroboration of H5's motivation; use "37% better recall via multi-index" only as an anecdotal, clearly-attributed datapoint.
- *Quote to extract*: the framing that chasing "perfect retrieval scores" is a trap.

4. "open-rag-eval: RAG Evaluation Without Golden Answers" — Ofer Mendelevitch, Vectara

- *Speaker/affiliation/event*: Ofer Mendelevitch, Vectara; AI Engineer. Built with University of Waterloo researchers (Jimmy Lin's lab).
- *Content*: Open-source, Apache-2.0 framework implementing TREC-RAG metrics — UMBRELA (LLM-judged passage relevance) and an AutoNuggetizer/nugget-based answer scoring — enabling evaluation without predefined golden answers. Two-step design: retrieve context+response per query via a connector, then run evaluators for retrieval and factual-consistency/hallucination metrics. Later added a Consistency-Adjusted Index measuring answer stability across repeated generations (relevant if you report reproducibility).
- *Maps to*: Evaluation (directly), H5, and your recent QA-generator quality fix — the most methodologically relevant talk for expanding your benchmark beyond deterministic qrels.
- *Academic grounding*: STRONG. UMBRELA and TREC-RAG are academic (Jimmy Lin lab). The most academically grounded evaluation talk in your list.
- *How to cite*: Citable methodological source in your Evaluation chapter; justify a reference-free evaluation arm alongside your MRR/nDCG qrels.
- *Data points*: UMBRELA + AutoNuggetizer metrics; reference-free two-stage architecture.

5. "HybridRAG: Fusion of Graph and Vector Retrieval" — Mitesh Patel, NVIDIA

- *Speaker/affiliation/event*: Mitesh Patel, NVIDIA; AI Engineer World's Fair 2025. Based on the BlackRock/NVIDIA paper "HybridRAG: Integrating Knowledge Graphs and Vector Retrieval Augmented Generation for Efficient Information Extraction" (Sarmah et al., arXiv:2408.04948).
- *Content*: Combines VectorRAG and GraphRAG (knowledge-graph triples built via LLM extraction, stored with LangChain's NetworkxEntityGraph). Evaluated on financial earnings-call transcripts. **Corrected figures from the paper's Table 5** (the numbers to quote): **context precision** — GraphRAG **0.96**, VectorRAG **0.84**, HybridRAG **0.79**; **context recall** — VectorRAG and HybridRAG **1.0**, GraphRAG **0.85**; **faithfulness** — GraphRAG and HybridRAG tie at **0.96** (VectorRAG 0.94); **answer relevancy** — HybridRAG **0.96** (VectorRAG 0.91, GraphRAG 0.89). The paper concludes HybridRAG "emerges as the most balanced and effective approach... in key metrics such as faithfulness and answer relevancy," while noting HybridRAG trades away context precision (0.79, the lowest of the three) to gain recall.
- *Maps to*: H3 (graph for entity/relation), H4 (hybrid fusion achieving highest aggregate — the closest published analog to your Hybrid hypothesis, though they fuse graph+**vector**, not graph+**lexical**), Literature Review, Methodology.
- *Academic grounding*: STRONG — direct peer-reviewed paper. Keystone citation.
- *How to cite*: Cite the arXiv paper as primary literature and the talk as its practitioner presentation; explicitly contrast your **vector-free** (BM25+graph) fusion against their graph+vector fusion as a differentiator. Their context-precision-vs-recall tradeoff is a ready-made explanatory frame for your H4 results.
- *Data points*: the Table 5 figures above (precision 0.96/0.84/0.79; recall 1.0/1.0/0.85; faithfulness 0.96/0.96/0.94; answer relevancy 0.96/0.91/0.89).

6. "When Vectors Break Down: Graph-Based RAG for Dense Enterprise Knowledge" — Sam Julien, Writer

- *Speaker/affiliation/event*: Sam Julien, Writer; published 22 July 2025 (AI Engineer World's Fair 2025).
- *Content*: Argues vector search fails on "dense mapping" — thousands of documents where similar terms repeat, so cosine similarity cannot discriminate. Writer's solution: a specialized text-to-graph model, graph structures stored as JSON in a Lucene-based search engine, and Fusion-in-Decoder (FiD) to integrate graph context — reducing hallucinations for regulated industries (healthcare, finance).
- *Maps to*: H3 (graph beats vectors on entity-dense corpora — highly relevant since welfare-scheme documents are terminology-dense with repeated entities), Literature Review, Limitations (vector failure modes).
- *Academic grounding*: Fusion-in-Decoder is published (Izacard & Grave 2021); the talk itself is practitioner/vendor. Medium.
- *How to cite*: Industry corroboration that vector similarity degrades on entity-dense corpora — a strong motivating parallel to your domain; pair with the FiD paper for the method.
- *Quote to extract*: the "dense mapping" failure-mode framing for vector search.

7. "GraphRAG Methods for Optimized LLM Context Windows" — Jonathan Larson, Microsoft Research

- *Speaker/affiliation/event*: Jonathan Larson, Senior Principal/Partner Data Architect, Microsoft Research; AI Engineer.
- *Content*: Microsoft GraphRAG pipeline — LLM entity/relationship extraction, community detection (Leiden), community-summary generation, and query-time assembly of graph structures into the context window. Demonstrated on the VIINA dataset (Russia-Ukraine news) where baseline vector RAG retrieved irrelevant text on global-sensemaking questions while GraphRAG surfaced coherent themes. The FastGraphRAG variant substitutes NLP (NLTK/spaCy noun-phrase extraction) for some LLM calls — directly analogous to your spaCy NER + NetworkX approach. [Microsoft](#) [GitHub](#)
- *Maps to*: H3 (multi-hop/global reasoning), Methodology (your GraphRAG design mirrors FastGraphRAG's NLP-based extraction), Literature Review.
- *Academic grounding*: STRONG — documented in Edge, Trinh, Cheng, Bradley, Chao, Mody, Truitt, Metropolitansky, Ness & Larson, "From Local to Global: A Graph RAG Approach to Query-Focused Summarization" (Microsoft Research, April 2024, arXiv:2404.16130), demonstrated "for a class of global sensemaking questions over datasets in the 1 million token range." Keystone citation.
- *How to cite*: Cite Edge et al. (2024) as primary literature for GraphRAG; note that your spaCy+NetworkX entity/chunk graph is a lightweight, vector-free cousin of FastGraphRAG.
- *Data points*: community-summary architecture; FastGraphRAG's NLP-substitution for cost reduction; the "global sensemaking" query class where vector RAG fails.

8. "How We Taught Agents to Use Good Retrieval" — Hanna Lichtenberg, Mixedbread AI

- *Speaker/affiliation/event*: Hanna Lichtenberg, Mixedbread AI; AI Engineer World's Fair 2026 (Search & Retrieval track).
- *Content*: Rebuts the "all you need is BM25/grep" claim (opening: "RAG is dead. Again. Vector search is useless. All you need is BM25. Not even BM25, all you need is grep."). Core thesis: retrieval models "falter because agents are not taught to leverage more sophisticated tools" — the "knowledge gap" between exponentially-improving LLM reasoning and incrementally-improving retrieval. Three reasons agents default to keyword/grep: (1) trained on coding tasks using grep, (2) trained on human-optimized web search, (3) **benchmark bias — BEIR/Nano-BEIR use entity-based queries that structurally favor BM25**. Presents a four-tool agent harness (overview search; main semantic search; filter-by-metadata; grep) with agent-selected tool per sub-query, max 4 rounds. Benchmarks (oracle-gap method): BrowseComp Plus (100k docs) oracle **93%** vs GPT-4 default **84%** (9-pt gap), reduced to **3 pts** with Mixedbread; an internal Treasury-docs QA oracle **64%** vs **56%**; an OBELICS/Congress benchmark **NDCG@10 0.4** vs a prior GPT-4 agent's **0.18**; and mgrep vs grep in Claude Code using **~2× fewer tokens** at similar/better quality. (AI-Weekly) (zenml)
- *Maps to*: H1 (BM25's real-but-overstated strength — she explains WHY BM25 looks so good, a crucial nuance for your H1 framing), H2 (semantic value when queries are well-formed), Phase-2 routing extension (agent tool-selection is the practitioner alternative to your trained router).
- *Academic grounding*: References BEIR/Nano-BEIR (Thakur et al. 2021) and public benchmarks; does NOT cite Adaptive-RAG/Self-RAG/CRAG/FLARE. Medium-strong.
- *How to cite*: Key industry evidence for H1's nuance (benchmark bias inflates BM25) and the leading example of "agentic tool-selection" you contrast against classifier-based routing in your Framework chapter.
- *Data points*: 93%→84% (9-pt) BrowseComp Plus oracle gap; benchmark-bias argument that BEIR queries favor BM25.

STRONG EXPERIMENTS & BENCHMARKS

9. "Evaluating AI Search: A Practical Framework" — Julia Neagu & Deanna Emery (Quotient AI), with Tavily

- *Speaker/affiliation/event*: Julia Neagu (CEO, Quotient AI; former Director of Data for GitHub Copilot), Deanna Emery (Founding AI Researcher, Quotient AI), plus a Tavily speaker; AI Engineer World's Fair (21-min talk). (Class Central)
- *Content*: Framework evaluating RAG/SAG/agents on three dimensions — source relevance to query, answer completeness, and faithful use of sources — using golden sets and LLM-as-judge with statistical rigor (confidence intervals, variance decomposition). Early benchmark comparisons focused on understanding where approaches fail rather than simple rankings.
- *Maps to*: Evaluation, H5 (faithfulness as a separate axis), your six query-type categorization approach.
- *Academic grounding*: Practitioner framework with statistical rigor; LLM-as-judge is published (Zheng et al. 2023). Medium.
- *How to cite*: Industry corroboration for separating relevance/completeness/faithfulness; useful methodological reference for your evaluation design.
- *Data points*: three-dimension framework; golden-set + LLM-judge + variance-decomposition methodology.

10. "Vector Search Benchmark[eting]" — Philipp Krenn, Elastic

- *Speaker/affiliation/event*: Philipp Krenn, Elastic; AI Engineer.
- *Content*: "Every vector database is both faster and slower than any other competitor — if you believe all the benchmarking." Deconstructs how vendors cherry-pick benchmark configurations; argues for honest, workload-relevant benchmarking (echoing Jim Gray's "benchmarking" critique and TPC history). (Typedef)
- *Maps to*: Limitations, Methodology (benchmark validity), directly relevant to your recent QA-generator bug fix and need for a clean benchmark.
- *Academic grounding*: Practitioner, but invokes the established "benchmarking" concept (Jim Gray). Medium.
- *How to cite*: Methodological caution when presenting your own latency/MRR comparisons — a useful hedge in your Limitations chapter.
- *Quote to extract*: "Every vector database is both faster and slower than any other competitor — if you believe all the benchmarking."

11. "Benchmarking Semantic Code Retrieval on Claude Code" — Kuba Rogut, Turbopuffer

- *Speaker/affiliation/event*: Kuba Rogut, Turbopuffer; AI Engineer (Europe/2026). (StartupHub.ai)
- *Content*: Quantifies wasted retrieval in Claude Code: "By default, Claude Code wastes one in every three file reads. Add windowed grep and that drops to one in five. Add semantic search on top and it drops to one [fewer still]." Argues semantic search meaningfully improves agent precision/efficiency over grep-only; early Claude Code used local vector RAG, then the team explored agentic search.
- *Maps to*: H1/H2 tradeoff (lexical vs semantic), Phase-2 (layering grep+semantic) — but domain is code, not documents.
- *Academic grounding*: Practitioner benchmark; no academic citations. Weak-medium.
- *How to cite*: Motivating anecdote for the complementarity of lexical + semantic retrieval; not a domain-matched citable source.
- *Data points*: one-in-three file reads wasted by default → one-in-five with windowed grep → fewer with semantic search.

12. "Navigating RAG Optimization With an Evaluation-Driven Compass" — Atita Arora (Qdrant) & co-presenter (Quotient AI)

- *Speaker/affiliation/event*: Atita Arora (Qdrant) and a Quotient AI co-presenter; AI Engineer.
- *Content*: Evaluation-driven RAG optimization — start with naive RAG, then add query expansion/rewriting and reranking/fusion, measuring at each step. Focus metrics: context relevance, chunk relevance (signal vs noise), and faithfulness (the hallucination metric). Demonstrates iterative, data-driven improvement rather than guesswork.
- *Maps to*: Evaluation, H5 (faithfulness separate from retrieval), Methodology (iterative optimization).
- *Academic grounding*: Practitioner/vendor; metrics align with RAGAS (Es et al. 2024). Medium.
- *How to cite*: Industry corroboration of evaluation-driven iteration; pair with the RAGAS paper for the underlying metrics.
- *Data points*: context relevance / chunk relevance / faithfulness triad.

13. "The RAG Stack We Landed On After 37 Fails" — Jonathan Fernandez

- *Speaker/affiliation/event*: Jonathan Fernandez, independent AI engineer building RAG for financial institutions; AI Engineer 2025. (zenml)
- *Content*: Journey from naive RAG through 37 failed implementations to a production stack. Lessons: prototyping tools ≠ production tools (closed APIs unacceptable for on-prem finance); a single test query is insufficient (use RAGAS). Final stack: LlamaIndex orchestration; Qdrant vector DB; open embeddings (BAAI BGE-small, NVIDIA); open LLMs (Llama 3.2, Qwen 3 4B) via Ollama/HuggingFace TGI at temperature 0; PII-removal preprocessing; bi-encoder first-stage retrieval then cross-encoder reranking (retrieve top 5 → rerank → pass top 2 to LLM); Arize Phoenix tracing; RAGAS evaluation; Docker-Compose microservices for on-prem deployment. (zenml + 2)
- *Maps to*: Methodology (the retrieve-top-5→rerank→top-2 pattern is a concrete, adoptable design for your Hybrid pipeline), Limitations (production pitfalls), your Framework's infrastructure-constraints axis (on-prem/privacy — directly relevant to a government-document domain).
- *Academic grounding*: Pure practitioner experience; no academic citations. Weak grounding, high practical value.
- *How to cite*: Practitioner case study for production pitfalls and the infrastructure-constraints dimension of your Framework (government/on-prem parallels finance); not an empirical citation for hypotheses.
- *Data points*: "37 fails"; retrieve top 5 → rerank → top 2; temperature 0; BGE-small/Qwen 3/ Llama 3.2 open stack.

SUPPORTING THEORY

14. "The Hidden Life of Embeddings" — Linus Lee, Notion

- *Speaker/affiliation/event*: Linus Lee, Research Engineer at Notion; AI Engineer Summit 2023 (San Francisco). (Aionlineclassroom)
- *Content*: Tour of embedding visualization/manipulation/interpretation — t-SNE, UMAP, PCA — plus using LLMs to manipulate embeddings and a tool observing which features popular embedding models encode. Emphasizes embeddings' life beyond vector search.
- *Maps to*: Literature Review / background on embeddings (contextualizes why FAISS + all-MiniLM behaves as it does), H2 (semantic embedding behavior).
- *Academic grounding*: References standard methods (t-SNE, UMAP, PCA); exploratory. Medium.
- *How to cite*: Accessible background on embedding interpretability; not a hypothesis-bearing source.
- *Quote to extract*: embeddings as more than "their value in vector search."

15. "State of AI-Powered Search and Retrieval"

- *Speaker/affiliation/event*: NOT reliably confirmed from available sources — likely a Weaviate/Cohere/Vespa-adjacent state-of-the-field talk, but speaker/date unverified. Verify on the YouTube page.
- *Content*: State-of-the-field survey of search/retrieval (embeddings, hybrid search, reranking, RAG evaluation), inferred from title and channel context.
- *Maps to*: Literature Review (landscape framing).
- *Academic grounding*: Unknown pending verification.
- *How to cite*: Verify speaker/affiliation first; if a vendor survey, use only as a landscape/motivation reference.

16. "How Codeium Breaks Through the Retrieval Ceiling"

- *Speaker/affiliation/event*: Codeium (now Windsurf) engineering team; speaker/date not reliably confirmed — verify.
- *Content*: Codeium's approach to code retrieval at scale — going beyond embedding-based retrieval ("retrieval ceiling") for coding agents. Domain is code, not documents.
- *Maps to*: Limitations (retrieval ceilings), tangential to your domain.
- *Academic grounding*: Practitioner/vendor. Weak.
- *How to cite*: Skip or use only as a passing anecdote — domain-mismatched (code) and thin on citable substance.

17. "Retrieval-Augmented Generation in the Wild"

- *Speaker/affiliation/event*: Not reliably confirmed from available sources — verify on the YouTube page.
- *Content*: Presumed practitioner survey of real-world RAG deployment challenges, inferred from title.
- *Maps to*: Limitations, Methodology (real-world deployment).
- *Academic grounding*: Unknown pending verification.
- *How to cite*: Verify before citing; likely a motivating-anecdote source rather than empirical.

The Adaptive-Routing Gap (your novelty check)

A targeted search of the AI Engineer 2023–2026 archives found **no talk whose central thesis is query-adaptive retrieval-paradigm routing** in the Adaptive-RAG sense (a per-query classifier choosing lexical vs dense vs graph, or routing by query complexity), and **no talk that cites the academic routing canon**. For precision in your Literature Review, the canonical routing paper is: **Adaptive-RAG (Jeong, Baek, Cho, Hwang & Park, NAACL 2024, pp. 7036–7050; arXiv:2403.14403)**, which "trains a T5-Large classifier on automatically derived complexity labels to route among three retrieval strategies" (non-retrieval, single-step, multi-step). Note the routing axis there is retrieval *depth/steps*, not retrieval *paradigm* — which is exactly your differentiation.

The closest adjacent AI Engineer talks frame adaptivity as **agentic tool-use**, not trained routing:

- **Jo Kristian Bergum (Hornet.dev, fmr. Vespa/Yahoo)** ran a WF2026 side event "Retrieval for Agents" and a Berlin Buzzwords talk "Agentic Retrieval: Building Self-Optimizing Search Systems," describing agents that "dynamically adjust the retrieval system itself" — agent-driven adaptation, not query-complexity routing (blog datapoint: median BrowseComp-Plus answer with GPT-5 + BM25 takes 24 search calls). Treat "self-optimizing" as vendor positioning, not rigorous novelty. (Berlin Buzzwords) (Hornet)
- **Hanna Lichtenberg (Mixedbread, #8)** — agent-selected tool per sub-query, explicitly not a trained classifier.
- Non-AI-Engineer contrast worth a footnote: **Neo4j NODES "Agentic GraphRAG" (Athulya Anil, UMass)** does explicit "failure-aware routing... adaptively routes queries between vector search and graph traversal" (HotpotQA EM + RAGAS) — routing appears at other venues but not on the AI Engineer stage. (Neo4j)

Implication: Your differentiation holds on all three axes. You can state, with sourced backing, that (a) practitioner discourse routes via agents at runtime, not via empirically-grounded per-query paradigm classifiers; (b) the academic routing lineage (Adaptive-RAG et al.) routes on retrieval depth, not paradigm; and (c) your work — routing on retrieval *paradigm*, grounded in empirically-validated H1-H3, in the Indian welfare-scheme domain — bridges and extends both. None of your listed talks overclaim routing as "novel/unprecedented," so there is no hype to flag within the list itself.

Recommendations (Phase 2 Plan)

(a) Implementing Prompt-RAG and Hybrid BM25+GraphRAG with RRF fusion:

1. Adopt Sarmah et al. (arXiv:2408.04948) as your primary methodological benchmark, but frame your BM25+GraphRAG fusion as **vector-free** — a novel contrast to their graph+vector fusion. Their Table 5 result (HybridRAG wins faithfulness 0.96 / answer relevancy 0.96 and recall 1.0 but posts the lowest context precision, 0.79) gives you a concrete, quantified hypothesis to test against H4.
2. Use RRF with **k = 60** and equal weighting as your fusion default (Cormack, Clarke & Büttcher, SIGIR 2009, which uses "the smoothing factor $k = 60$ "), endorsed practitioner-side by Krenn; log the latency cost explicitly to test H4's "highest MRR at cost of higher latency" clause.
3. Borrow Fernandez's **retrieve-top-5 → rerank → top-2** pattern and temperature-0 generation for deterministic, auditable answers — important for a government-document domain with explainability requirements.
4. Treat Lichtenberg's four-tool agent harness as the "agentic baseline" to contrast against your rules-based router in Phase 2, so you can show your empirically-grounded router versus black-box agent tool-selection.

(b) Expanding your corpus and QA benchmark:

1. Add a reference-free evaluation arm using Vectara's open-rag-eval (UMBRELA + AutoNuggetizer) alongside your MRR/nDCG qrels — this directly addresses your QA-generator quality concerns and lets you evaluate without hand-authored golden answers as you scale beyond 120 documents.
2. Heed Lichtenberg's benchmark-bias warning: because entity-based queries structurally favor BM25 (as in BEIR), ensure your paraphrase-semantic query category (H2) uses genuinely divergent vocabulary, or you will under-detect FAISS's advantage and bias your H1/H2 comparison toward lexical.
3. Apply Krenn's "benchmarking" caution and Quotient AI's statistical rigor (confidence intervals, variance decomposition) when reporting cross-system comparisons — pre-register your six query-type splits before the re-evaluation run.

(c) Writing the final results chapter:

1. Keep retrieval metrics and generation faithfulness in separate sections (H5) — cite the AI21 "RAG Evaluation Is Broken" talk and the Quotient/Qdrant evaluation talks as industry corroboration that high retrieval scores can mislead.
2. Build the Literature Review's graph-RAG section on the three strongly-grounded talks (NVIDIA/Sarmah et al., Microsoft/Edge et al., Writer/FiD) plus their papers; use practitioner talks as "industry corroboration" sentences, not load-bearing claims.
3. In the Framework chapter, position your contribution against the confirmed gap: practitioner conferences route via agents; academia (Adaptive-RAG) routes via a trained classifier on retrieval *depth*; your framework routes on retrieval *paradigm*, grounded in empirically-validated H1-H3, in a novel domain.

Benchmarks that would change these recommendations: If your re-run shows BM25 does NOT beat FAISS on terminology-sensitive queries (H1 fails), downgrade the lexical-comeback talks to background and re-center on the semantic cluster. If Hybrid does not top aggregate MRR (H4 fails), the NVIDIA context-precision-vs-recall tradeoff becomes your explanatory frame. If GraphRAG underperforms on multi-hop (H3 fails), foreground counter-literature such as Samsung's "UnWeaving the knots of GraphRAG — turns out VectorRAG is almost enough" (arXiv:2603.29875).

Caveats

- Videos #15, #16, #17 could not have their speakers/dates/content reliably confirmed from available sources; verify these directly on their YouTube pages before citing. Treat all three as motivating-anecdote sources pending confirmation.
- Talk content for #1, #8, and #13 was partly reconstructed from official descriptions and high-quality secondary write-ups (e.g., ZenML's LLMOps Database) because direct YouTube transcript fetching was rate-limited; the Mixedbread (#8) and Fernandez (#13) details are well-sourced, but treat the Krenn (#1) specifics as description-plus-known-canon rather than verbatim in-talk quotes.
- **Fact-check correction incorporated:** an earlier draft mis-stated the HybridRAG figures; the paper's Table 5 shows GraphRAG context *precision* 0.96 (not ~0.85) and GraphRAG context *recall* 0.85 — the corrected numbers are used above. Verify against the paper's Table 5 before quoting in your dissertation.
- The adaptive-routing gap is a negative result across catalog listings, track summaries, and searches; it cannot prove no such talk exists in every unlisted session, but no evidence of a query-complexity/paradigm-selection router talk (or citation of Adaptive-RAG/Self-RAG/CRAG/FLARE) surfaced.
- Several vendor talks (Writer, Qdrant, Vectara, Mixedbread, NVIDIA, Elastic) have commercial interests; treat their performance claims as directional and always pair with the underlying peer-reviewed method where one exists.